■ DevPilot AI — Frontend Architecture & Design Spec

Official Technical Document for Multi-Agent UI System (v1.0)

---

1. Technology Stack & Design System

Framework: Next.js 15 (App Router) with React 19, TypeScript, and Tailwind CSS.

Color Palette: Primary Navy (#0F172A), Accent Indigo (#6366F1), Dark Mode Background (#020617), Card Border (#1E293B).

State Management: Zustand for global app state & authentication tokens. TanStack Query v5 for server state caching.

2. Core Multi-Agent UI Components

• **AgentSelector.tsx:** Interactive dropdown enabling real-time switching between Planner, RAG, and GitHub agents.

• **StreamMessage.tsx:** Server-Sent Events (SSE) consumer rendering markdown responses token-by-token with glowing cursor animations.

• **DocumentUploader.tsx:** Drag-and-drop file upload zone accepting .pdf, .md, .txt, and .docx files up to 10MB.

• **TaskPlanTree.tsx:** Collapsible task hierarchy component displaying sub-tasks created by the Planner Agent.

3. API Gateway & Security Specifications

• **Authentication:** JWT Bearer Tokens issued via POST /api/v1/auth/login and stored securely in localStorage.

• **RAG Upload Endpoint:** POST /api/v1/knowledge/documents/upload accepting multipart form data.

• **Rate Limiting:** API limits clients to a maximum of 100 requests per minute per IP address.

4. Recommended Sample Queries for RAG Testing

1. *'What color tokens are used in the DevPilot design system?'*

2. *'Which component handles token-by-token streaming markdown in the UI?'*

3. *'What are the allowed file types and size limits for document uploads?'*